16 – Intro. to Pricing

IEOR 145/245

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So far …

Engineering demand by:

- Allocating resources with constraints and uncertain demand
  - When customers don’t choose

- Finding good set of options to show
  - When customers choose

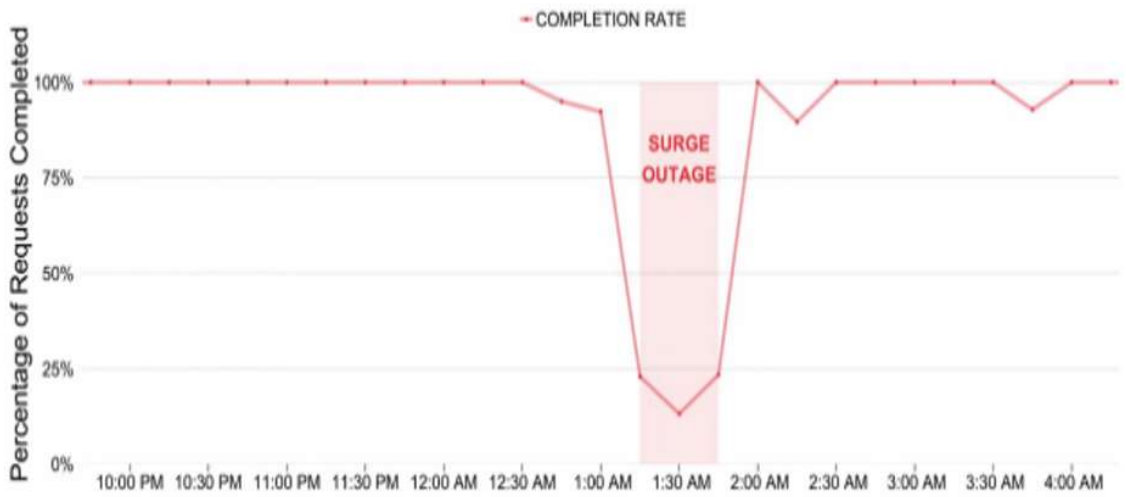
This Module (# 3)

Engineering demand by:

- Allocating resources with constraints and uncertain demand
  - When customers don't choose
- Finding good set of options to show
  - When customers choose
- **Pricing:** when allocations are indirectly controlled through prices

Impact of pricing in Ridesharing

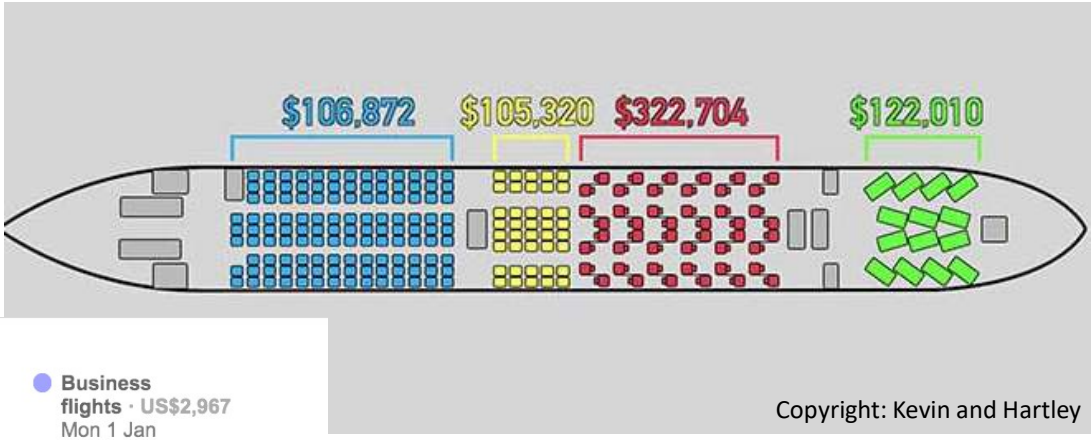
(NYC during 2014-15 New Year's Eve)



Srce: Hall, Kendrick and Nasko, U. of Chicago

Pricing in Airline Seat Reservation

- How many seats in each class?
- At what price?



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Effect of Pricing on Demand

1. If price is too high, demand is lower

2. If price is very low, demand is higher

- How do we formally model this?

- How do we decide optimal price?

Prices from Customer Viewpoint

Value or Willingness to Pay

- Suppose you are hungry and there is just 1 option – V&A cafe
- Knowing about lack of options, V&A keeps a high price
- How much will you pay to eat?
  1. 1 million USD
  2. 401k USD
  3. 30 USD
  4. 0 USD

Value or Willingness to Pay

- Suppose you are hungry and there is just 1 option – V&A cafe
- Knowing about lack of options, V&A keeps a high price
- How much will you pay to eat?
- Perhaps not more than $30
  - The highest you are willing to pay is your **value or willingness to pay**

Utility

Net utility of purchase = value/willingness to pay – price ≥ 0

- Customer purchases if *value (in unit of price) exceeds the price*

Value of a Population

- Sometimes we don't know how much we value an item
  - How much do we need the item?
  - Item may have emotional value
  - Peer pressure
  - …
  - So how can a retailer know this?

- Individual values are hard but population's value is easier
  - How? – **Distribution over values**

- **E.g.,** a population with uniformly distributed value on a scale of {1, 2, 3, …, 10}
  - $\frac{1}{10}$ th of population has value v, $\forall v \in \{1, 2, \dots, 10\}$

General Model of Value

- A probability distribution F over values
  - $F(x)$ is the fraction of population with value $< x$

- Given a price p, what is the fraction of population willing to pay?

General Model of Value

- A probability distribution F over values
  - $F(x)$ is the fraction of population with value $< x$

- Given a price p, fraction $1 - F(p)$ of population willing to pay
  - These are customers with *value at least* p

- Alternatively, probability $1 - F(p)$ that a random customer buys the item

Properties of Model

- A probability distribution F over values
  - $F(x)$ is the fraction of population with value $< x$
- Probability $1 - F(p)$ that a random customer buys the item

1. $1 - F(p)$ is monotonically non-increasing in p
2. As $p \to +\infty$, goes to 0
3. As $p \to 0$, goes to 1

Prices from Seller Viewpoint

What Does a Seller Want?

- To maximize profit
  - Set high prices?
- To sell all items
  - Set low prices?
- Is any price OK?
  - Sellers have a min/max price - depends on the cost per product
  - Brand reputation
  - Supply chain contracts with upstream agents …

Seller's criteria: Feasibility Set

- Set $\mathcal{F}$ of prices are feasible

- $\mathcal{F}$ could just be lower and upper bounds $\mathrm{p} \in [p_{min}, p_{max}]$

Optimal Pricing

Optimal Price

- E.g., a population with uniformly distributed value on a scale of {1, 2, 3, …, 10}
- Seller finds all prices in the range [1,10] feasible
- What is the price that maximizes:
  - # of units sold?
  - revenue?

Optimal Price

- E.g., a population with uniformly distributed value on a scale of {1, 2, 3, …, 10}

- Seller finds all prices in the range [1,10] feasible

- What is the price that maximizes:
  - # of units sold? – price = 1
  - revenue? – price = 5

Sales Maximizing Price

- Set price to *minimum feasible* price for seller
- Maximizes # customers who buy
- May lead to low profit
  - But customers are very happy

Revenue Optimal Pricing

Revenue function: $R(p) = p\left(1 - F(p)\right)$

(typically, concave function)

- $\left(1 - F(p)\right)$ is the probability that a random customer has value $\geq p$

- **Optimal price:**

$$p^* = \underset{p \in \mathcal{F}}{\operatorname{argmax}}\, R(p)$$

Revenue Optimal Pricing

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- **Optimal price:** $p^* = \underset{p \in \mathcal{F}}{\operatorname{argmax}}\, R(p)$

Example: $F(p) = p, \mathcal{F} = [0.2, 0.8]$ – what is optimal price?

Implicit assumptions?

- Can I deploy this in reality?

Implicit assumptions

1. We know distribution $F(x)$
   - **Reality:** Don't know this and it depends on competition

Implicit assumptions

1. We know distribution $F(x)$

2. We have infinite inventory of product
   - **Reality:** typically have a finite amount

Implicit assumptions

1. We know distribution $F(x)$

2. We have infinite inventory of product

3. There is just one product with variable price
   - **Reality:** many products with substitution (and complementarity)
   - Changing price of one product affects demand for many others

Implicit assumptions

1. We know distribution $F(x)$

2. We have infinite inventory of product

3. There is just one product with variable price

4. We have no information about customer on arrival
   - **Reality:** We do have some knowledge (e.g., time)
   - **Price differentiation**

Landscape of Pricing Problems

1. We know/don't know distribution $F(x)$
2. We have finite/infinite inventory of product
3. There are many/one products
4. We have information/no info. about customer on arrival

Pricing a Single Item with Unknown Demand

Revenue Optimal Pricing

Revenue function: $R(p) = p\left(1 - F(p)\right)$

(typically, concave function)

- Optimal price: $p^* = \underset{p \in \mathcal{F}}{\operatorname{argmax}}\, R(p)$

- **How do we optimize without knowing F?**

Three Approaches

1. **Learn from samples**

2. Experiment based on small price changes

3. Use similarity to a product with known distribution

1. Learning from samples

- **Data:** prices and corresponding number of units sold $\{p_t, N_t\}_{t \in T}$
  - When price was p_t, N_t units were sold

- **Goal:** find optimal price i.e., solve $\max_p R(p)$

1. Learning from samples

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Idea:

1. **Assume the value distribution has a specific form**

   Examples:
   - $F(p) = 1 - e^{-\alpha p}$ (α is unknown)
     - **Why?:** Price change from 0 to 1 is more significant than 100 to 101
   - $F(p) = \min\left\{1, \frac{p}{\alpha}\right\}$ (α is unknown)
     - **Why?:** α is competitor's price

2. **Find a good fit using data and optimize price**

How do we find the fit from data?

- Data: prices and corresponding number of units sold $\{p_t, N_t\}_{t \in T}$
  - When price was p_t, N_t units were sold

- **Goal:** Learn $F(p)$

- **Simplification:** Suppose we have data $\{p_t, F(p_t)\}_{t \in T}$

How do we find the fit from data?

Simplification: Suppose we have data $\{p_t, F(p_t)\}_{t \in T}$

For illustration: Let $F(p) = \min\left\{1, \frac{p}{\alpha}\right\}$ (α is unknown)

Two steps: regression then optimization

1. *Finding fit:* $\alpha^* = \operatorname{argmin}_\alpha \sum_t \left(\min\left\{1, \frac{p_t}{\alpha}\right\} - F(p_t)\right)^2$ (Regression)

2. *Optimal price:* $p^* = \operatorname{argmax}_p \; p(1 - F(p))$

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Is this the best approach?

Fit with Decision in Mind (End-to-end learning)

Simplification: Suppose we have data $\{p_t, F(p_t)\}_{t \in T}$

For illustration: Let $F(p) = \min\left\{1, \frac{p}{\alpha}\right\}$ (α is unknown)

1. *Finding fit for* $R(p)$ (instead of $F(p)$)*:*

$$\alpha^* = \operatorname{argmin}_\alpha \sum_t \left(p_t \min\left\{1, \frac{p_t}{\alpha}\right\} - p_t F(p_t)\right)^2$$

2. *Optimal price:* $p^* = \operatorname{argmax}_p \; p(1 - F(p))$

Which approach is better?

- Regression then optimization or end-to-end learning?

- It depends on the quality of the data
  - If data is unbiased (equally noisy at all prices): end-to-end is better

  - If data is more noisy for higher prices: end-to-end can be worse

Back to actual data

Simplification: Suppose we have data $\{p_t, N_t\}_{t \in T}$
(instead of $\{p_t, F(p_t)\}_{t \in T}$)

What do we do?

Back to actual data

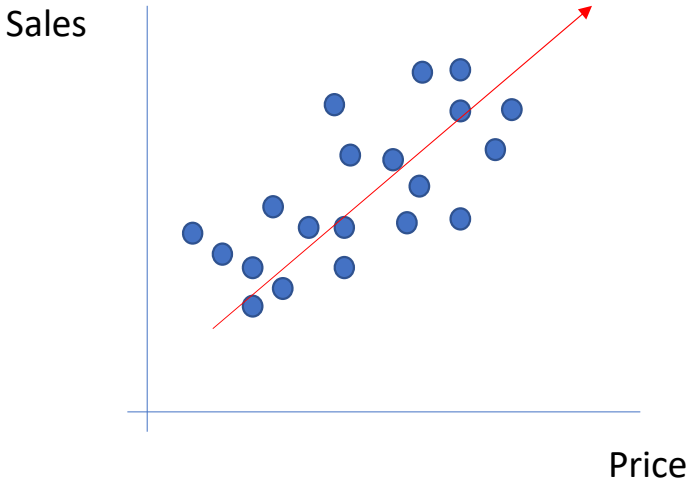
Simplification: Suppose we have data $\{p_t, N_t\}_{t\in T}$
(instead of $\{p_t, F(p_t)\}_{t\in T}$)

- Learn the function $NF(p)$ instead of $F(p)$

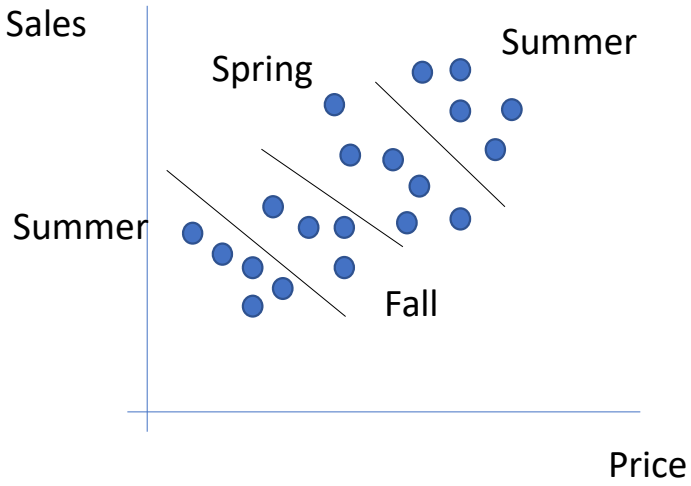
$$\text{Learning step} = \operatorname{argmin}_{N,\alpha} \sum_t \left(N \min\left\{1, \frac{p_t}{\alpha}\right\} - N_t\right)^2$$

(two unknowns: harder optimization problem)

Beware of Confounders



Beware of Confounders



Need to neutralize confounding factors
- Human expertise
- ML techniques

Three Approaches

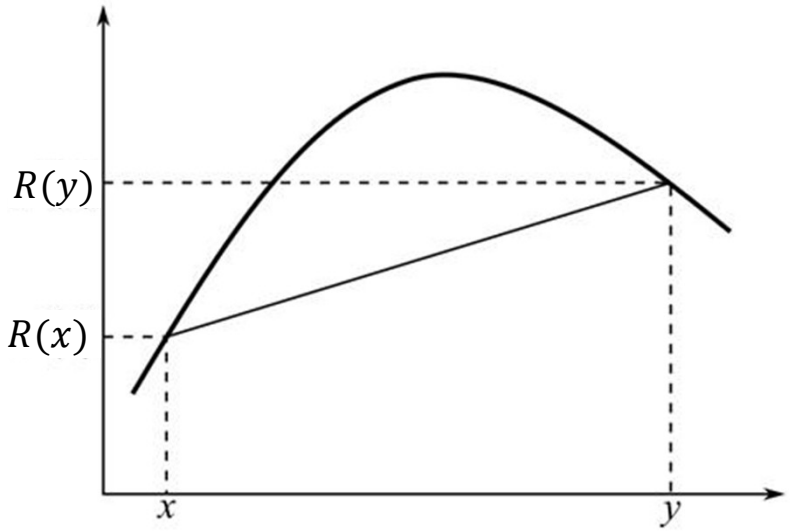
1. Learn from samples
   - Covered last lecture

2. Experiment based on small price changes

3. Use similarity to a product with known distribution

2. Experiment based on price changes

Recall: For typical distributions F, revenue $R(p)$ is concave function



2. Experiment based on price changes

Recall: For typical distributions F, revenue $R(p)$ is concave function

Consider a small change in price:

1. Given current price p_t, experiment with $p_{t+1} = p_t(1 + \epsilon)$
   - Let new number of units sold be N_{t+1}

2. Experiment based on price changes

Recall: For typical distributions F, revenue $R(p)$ is concave function

Consider a small change in price:

1. Given current price p_t, experiment with $p_{t+1} = p_t(1+\epsilon)$
   - Let new number of units sold be N_{t+1}
2. Compute revenue difference: $p_{t+1}\,N_{t+1} - p_t N_t$

2. Experiment based on price changes

Recall: For typical distributions F, revenue $R(p)$ is concave function

Consider a small change in price:

1. Given current price p_t, experiment with $p_{t+1} = p_t(1 + \epsilon)$
   - Let new number of units sold be N_{t+1}
2. Compute revenue difference: $p_{t+1}\, N_{t+1} \; - p_t N_t$
   - If > 0, repeat step 1 (i.e., increase price further)
   - If < 0, experiment with lower price next i.e.,

$$p_{t+2} = p_t(1 - \epsilon)$$

2. Experiment based on price changes

- Increase price a little from current value (demand will decrease)
  1. If net profit increases, let $p_{t+1} = (1 + \epsilon)\, p_t$
  2. If net profit decreases, let $p_{t+1} = (1 - \epsilon)\, p_t$

- Stop when price change reverses direction a few times
  - Alternatively, when change is revenue starts to become “very small”
  - Why? – concavity of revenue function

What is a good value of ϵ ?

- Large value may miss optimum
- Small value requires too much experimentation
  - Impractical
- Right tradeoff is a business/contextual decision

Three Approaches

1. Learn from samples

2. Experiment based on small price changes

3. **Use similarity to a product with known distribution**

3. Use similarity to another product

- Suppose we have found the optimal price of a “similar” product in the past
- Use the same (or similar) revenue function
- How would you define “similar”?
  - Use product features and perform **clustering** in feature space (ML software)

Other approaches?

- Many more…

- Main ideas are similar, often a combination of ideas used in practice